King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam I Term 243

Original exam date: 3 July 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The value of k for which the vectors $\mathbf{u} = (1, 0, 2)$, $\mathbf{v} = (-1, 2, k)$ and $\mathbf{w} = (0, 9, 3)$ of $\mathbb{R}^3$ are linearly dependent is

(a) $-\frac{4}{3}$

(b) $\frac{4}{5}$

(c) $\frac{3}{4}$

(d) $-\frac{2}{3}$

(e) $\frac{2}{3}$

2. Let $\mathbf{v} = (2, 5, -4)$ and $\mathbf{w} = (1, -2, -3)$ be two vectors in $\mathbb{R}^3$. Find $|\mathbf{v} - 2\mathbf{w}|$

(a) $\sqrt{87}$

(b) $\sqrt{85}$

(c) 9

(d) $\sqrt{83}$

(e) $\sqrt{17}$

3. If the differential equation

$$\left(5x^4 - 4x^3y^3\right) dx + \left(5 + kx^4y^2\right) dy = 0$$

is an exact differential equation, then $k =$

(a) -6

(b) -5

(c) -4

(d) -7

(e) -3

4. If the acceleration of a particle is $a(t) = 18\cos(3t)$, the initial velocity is $v_0 = 4$, and the initial position is $x_0 = -7$, then $x\left(\frac{\pi}{2}\right) =$

(a) $2\pi - 7$

(b) $7 - 2\pi$

(c) $2\pi + 5$

(d) $2\pi + 7$

(e) $2\pi - 5$

5. By making a suitable substitution, the differential equation $xy' = y + \sqrt{x^2 + y^2}$ can be transformed into a separable differential equation

(a) $\dfrac{1}{\sqrt{1+v^2}}\,dv = \dfrac{1}{x^2}\,dx$

(b) $\sqrt{1+v^2}\,dv = x\,dx$

(c) $\dfrac{1}{\sqrt{1+v^2}}\,dv = x^2\,dx$

(d) $\dfrac{1}{\sqrt{1+v^2}}\,dv = \dfrac{1}{x}\,dx$

(e) $\sqrt{1+v^2}\,dv = \dfrac{1}{x}\,dx$

6. The solution space of the system

$$\begin{aligned} x_1 - 2x_2 + 2x_3 + x_4 &= 0 \\ x_1 - x_2 + 2x_3 + 5x_4 &= 0 \\ x_1 - 4x_2 + 2x_3 - 7x_4 &= 0 \end{aligned}$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$, where s and t are real numbers and

(a) $\mathbf{u} = (2, 0, 1, 0)$ and $\mathbf{v} = (9, -4, 0, 1)$

(b) $\mathbf{u} = (-2, 0, 1, 0)$ and $\mathbf{v} = (-9, -4, 0, -1)$

(c) $\mathbf{u} = (-2, 0, 1, 0)$ and $\mathbf{v} = (-9, -4, 0, 1)$

(d) $\mathbf{u} = (-2, 0, 1, 0)$ and $\mathbf{v} = (9, 4, 0, 1)$

(e) $\mathbf{u} = (-2, 0, -1, 0)$ and $\mathbf{v} = (9, -4, 0, 1)$

7. Let $\mathbf{u} = \langle 2, 3\rangle$, $\mathbf{v} = \langle -4, 5\rangle$, $\mathbf{w} = \langle -16, 42\rangle$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a + b =$

(a) 9

(b) 6

(c) 10

(d) 4

(e) 8

8. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

(a) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 = 1$

(b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 = 0$

(c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 x_3 = 0$

(d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 x_2 = 0$

(e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 = 1$

9. A general solution of the exact differential equation

$$\big(e^{2y} - y\cos(xy)\big)\,dx + \big(2xe^{2y} - x\cos(xy) + 2y\big)\,dy = 0$$

is

(a) $3xe^{2y} + y\sin(xy) + y^2 = c$

(b) $xe^{2y} + y\sin(xy) + y^2 = c$

(c) $xe^{2y} - \sin(xy) + y^2 = c$

(d) $2xe^{2y} - \sin(xy) + y^2 = c$

(e) $xe^{2y} - x\sin(xy) + y^2 = c$

10. By making a suitable substitution, the differential equation $x^2y' + 2xy = 5y^4$ can be transformed into a linear differential equation

(a) $v' - \dfrac{6}{x}v = -\dfrac{15}{x^2}$

(b) $v' - \dfrac{3}{x}v = -\dfrac{12}{x^2}$

(c) $v' + \dfrac{6}{x}v = -\dfrac{14}{x^2}$

(d) $v' - \dfrac{6}{x}v = -\dfrac{20}{x^2}$

(e) $v' + \dfrac{6}{x}v = -\dfrac{10}{x^2}$

11. The sum of all values of r such that $y = x^r$ is a solution of the differential equation

$$3x^2y'' + 6xy' + y = 0$$

is

(a) -5

(b) -4

(c) -1

(d) -2

(e) -3

12. A general solution of the differential equation $y' = (9x + 4y)^2$ is given by

(a) $y = \dfrac{1}{8}\left[\tan(6x + c) - 6x\right]$

(b) $y = \dfrac{3}{8}\left[\tan(6x + c) - 6x\right]$

(c) $y = \dfrac{1}{5}\left[\tan(6x + c) - 6x\right]$

(d) $y = \dfrac{1}{8}\left[\tan(6x + c) + 6x\right]$

(e) $y = \dfrac{3}{5}\left[\tan(6x + c) + 6x\right]$

13. A culture initially has P_0 number of bacteria. At $t = 1$ hour the number of bacteria is measured to be $\dfrac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t, the time necessary for the number of bacteria to triple is

(a) $\dfrac{\ln 3}{\ln 3 - \ln 2}$

(b) $\dfrac{\ln 3}{\ln 4 - \ln 3}$

(c) $\dfrac{\ln 3}{\ln 3 + \ln 2}$

(d) $\dfrac{\ln 2}{\ln 5 - \ln 4}$

(e) $\dfrac{\ln 3}{\ln 7 - \ln 5}$

14. The solution of the initial-value problem

$$\left(x^2 - 9\right)\frac{dy}{dx} = 12y, \qquad y(0) = 4$$

is

(a) $y = -4\left(\dfrac{x-3}{x+3}\right)^3$

(b) $y = 4\left(\dfrac{x-3}{x+3}\right)^4$

(c) $y = 4\left(\dfrac{x-2}{x+2}\right)^2$

(d) $y = -4\left(\dfrac{x-2}{x+2}\right)^3$

(e) $y = 4\left(\dfrac{x-3}{x+3}\right)^2$

15. The general solution of the linear differential equation $xy' = 2y + x^3 \cos x$ is given by

(a) $y(x) = x^3(\cos x + c)$

(b) $y(x) = x^2(\cos x + c)$

(c) $y(x) = x^3(\sin x + c)$

(d) $y(x) = x(\sin x + c)$

(e) $y(x) = x^2(\sin x + c)$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\det = 1(6-9k)+2(-9) = -12-9k = 0$, so $k = -\frac{4}{3}$.
2	**(b)**	$\mathbf{v}-2\mathbf{w} = (0,9,2)$, length $\sqrt{81+4} = \sqrt{85}$.
3	**(e)**	$M_y = -12x^3y^2$, $N_x = 4kx^3y^2$, so $4k = -12$ and $k = -3$.
4	**(e)**	$v = 6\sin 3t + 4$, $x = -2\cos 3t + 4t - 5$, so $x(\frac{\pi}{2}) = 2\pi - 5$.
5	**(d)**	Homogeneous, $v = y/x$: $xv' = \sqrt{1+v^2}$, so $\frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$.
6	**(c)**	RREF rows $(1,0,2,9)$, $(0,1,0,4)$: $s(-2,0,1,0)+t(-9,-4,0,1)$.
7	**(c)**	$2a-4b=-16$, $3a+5b=42$: $a=4$, $b=6$, so $a+b=10$.
8	**(b)**	Only $x_1 = 0$ (homogeneous linear) gives a subspace; the others fail sums or miss $\mathbf{0}$.
9	**(c)**	$F = \int M\,dx = xe^{2y} - \sin(xy) + g(y)$, $g' = 2y$: $xe^{2y} - \sin(xy) + y^2 = c$.
10	**(a)**	Bernoulli $n=4$, $v = y^{-3}$: multiply by $-3y^{-4}$ to get $v' - \frac{6}{x}v = -\frac{15}{x^2}$.
11	**(c)**	$3r^2+3r+1=0$, $r = -\frac{1}{2} \pm \frac{\sqrt{3}}{6}i$, so the sum is -1.
12	**(b)**	$v = 9x+4y$: $v' = 9+4v^2$, $v = \frac{3}{2}\tan(6x+c)$, $y = \frac{3}{8}[\tan(6x+c) - 6x]$.
13	**(a)**	$P = P_0e^{kt}$, $k = \ln\frac{3}{2}$; $e^{kt} = 3$ gives $t = \frac{\ln 3}{\ln 3 - \ln 2}$.
14	**(e)**	$\frac{dy}{y} = (\frac{2}{x-3} - \frac{2}{x+3})dx$: $y = K(\frac{x-3}{x+3})^2$, $K = 4$.
15	**(e)**	$y' - \frac{2}{x}y = x^2\cos x$, $\rho = x^{-2}$: $(x^{-2}y)' = \cos x$, $y = x^2(\sin x + c)$.